SUPPLEMENTAL MATERIAL

Table S1. Full search strategy.

Ovid MEDLINE – ALL (1946 to June 12th, 2020)
Searched on June 12th, 2020
RCT Filter: BMJ Publishing Group Limited. BMJ Best Practice Study design search filters 2017
Available from: https://bestpractice.bmj.com/info/us/toolkit/learn-ebm/study-design-search-filters/

1	*"Peer Review"/ or *"Peer Review, Research"/
2	(peer adj3 (review or reviewed or reviewing or reviewer or reviewers)).ti.
3	(blind review or blind reviewed or referee* or post-publication review or cascading review or third party review or author suggested reviewers or editor suggested reviewers or manuscript reviewer*).ti.
4	or/1-3
5	"randomized controlled trial".pt.
6	(random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab.
7	(retraction of publication or retracted publication).pt.
8	or/5-7
9	(animals not humans).sh.
10	((comment or editorial or meta-analysis or practice-guideline or review or letter) not "randomized controlled trial").pt.
11	(random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not "randomized controlled trial".pt.
12	8 not (9 or 10 or 11)
13	4 and 12

Table S2. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials.

	RANDOM SEQUENCE GENERATION	ALLOCATION CONCEALMENT	BLINDING OF PARTICIPANTS	BLINDING OF OUTCOME ASSESSMENT	INCOMPLETE OUTCOME DATA	SELECTIVE REPORTING	OTHER SOURCES OF BIAS
Alam, 2011	+	?	-	+	+	?	?
Arnau, 2003	+	+	+	+	+	?	?
Callaham, 2002	+	?	+	+	+	?	?
Callaham, 2002 JAMA	+	?	+	+	+	?	?
Cobo, 2011	+	+	+	+	+	+	?
Cobo, 2007	+	?	+	+	+	?	?
Das Sinha, 1999	+	?	-	+	+	?	?
Fisher, 1994	+	?	-	+	+	?	?
Godlee, 1998	+	?	-	+	+	?	?
Houry, 2012	+	?	?	+	+	?	?
John, 2019	+	?	+	+	+	?	?
Johnston, 2007	+	?	+	+	+	?	?
Justice, 1998	+	?	-	+	+	?	?
McNutt, 1990	+	?	-	+	+	?	?
Neuhauser, 1989	+	?	?	?	+	?	?
Okike, 2016	+	?	-	?	+	+	?
Pitkin, 2002	+	?	?	+	+	?	?
Provenzale, 2020	?	?	?	+	+	?	?
Schroter, 2004	+	?	-	+	+	?	?
Van Rooyen, 2010	+	?	+	+	+	?	?
Van Rooyen, 1998	+	?	-	+	+	?	?
Van Rooyen, 1999	+	?	?	+	+	?	?
Vinther, 2012	+	?	?	+	+	?	?
Walsh, 2000	+	?	?	+	+	?	?
	+	Low Risk					
	?	Uncertain					
	-	High Risk					

Table S3. Netsplit for the different peer-review interventions for the main outcome of peer-review quality (random effects model). Example: 19 studies compared standard peer-review process vs reviewer-level interventions and the estimated treatment effect (standardized mean difference [SMD]) was -0.198.

Random effects model:

comparison	k	prop	nma	direct	indir.	Diff	z	p-value
Author-level:Control	9	1.00	0.0958	0.0958	.	.	.	.
Author-level:Editor-level	0	0	0.0862	.	0.0862	.	.	.
Author-level:Reviewer-level	0	0	-0.1022	.	-0.1022	.	.	.
Control:Editor-level	5	1.00	-0.0096	-0.0096	.	.	.	.
Control:Reviewer-level	19	1.00	-0.1980	-0.1980	.	.	.	.
Editor-level:Reviewer-level	0	0	-0.1884	.	-0.1884	.	.	.

Legend:

- comparison - Treatment comparison
- k - Number of studies providing direct evidence
- prop - Direct evidence proportion
- nma - Estimated treatment effect (SMD) in network meta-analysis
- direct - Estimated treatment effect (SMD) derived from direct evidence
- indir. - Estimated treatment effect (SMD) derived from indirect evidence
- Diff - Difference between direct and indirect treatment estimates
- z - z-value of test for disagreement (direct versus indirect)
- p-value - p-value of test for disagreement (direct versus indirect)

Table S4. Quantifying heterogeneity/inconsistency, tests of heterogeneity (within designs) and inconsistency (between designs), and design-specific decomposition of within-designs Q statistic (main outcome of peer-review quality).

```
                     Q df  p-value
Total           130.42 30 < 0.0001
Within designs  130.42 30 < 0.0001
Between designs   0.00  0       --

Design-specific decomposition of within-designs Q statistic

                 Design      Q df  p-value
   Author-level:Control  17.57  8   0.0247
   Control:Editor-level   9.18  4   0.0568
 Control:Reviewer-level 103.68 18 < 0.0001

Q statistic to assess consistency under the assumption of
a full design-by-treatment interaction random effects model

                   Q df p-value tau.within tau2.within
Between designs 0.00  0      --     0.2525      0.0638
```

Table S5. League table for the main outcome of peer-review quality (fixed effect model).

Reviewer-level			
0.07 [-0.04; 0.17]	Author-level		
0.11 [-0.13; 0.36]	0.05 [-0.21; 0.30]	Editor-level	
0.14 [0.08; 0.20]	0.07 [-0.01; 0.16]	0.03 [-0.21; 0.27]	Control

Table S6. Netsplit for the different peer-review interventions for the main outcome of peer-review quality (fixed effect model). Example: 19 studies compared standard peer-review process vs reviewer-level interventions and the estimated treatment effect (standardized mean difference [SMD]) was -0.1397.

Fixed effect model:

comparison	k	prop	nma	direct	indir.	Diff	z	p-value
Author-level:Control	9	1.00	0.0738	0.0738	.	.	.	.
Author-level:Editor-level	0	0	0.0460	.	0.0460	.	.	.
Author-level:Reviewer-level	0	0	-0.0659	.	-0.0659	.	.	.
Control:Editor-level	5	1.00	-0.0278	-0.0278	.	.	.	.
Control:Reviewer-level	19	1.00	-0.1397	-0.1397	.	.	.	.
Editor-level:Reviewer-level	0	0	-0.1120	.	-0.1120	.	.	.

Legend:

comparison	Treatment comparison
k	Number of studies providing direct evidence
prop	Direct evidence proportion
nma	Estimated treatment effect (SMD) in network meta-analysis
direct	Estimated treatment effect (SMD) derived from direct evidence
indir.	Estimated treatment effect (SMD) derived from indirect evidence
Diff	Difference between direct and indirect treatment estimates
z	z-value of test for disagreement (direct versus indirect)
p-value	p-value of test for disagreement (direct versus indirect)

Table S7. League table for the duration of the peer-review process (random effects model).

Reviewer-level			
-0.02 [-0.39; 0.34]	Author-level		
-0.05 [-0.66; 0.57]	-0.02 [-0.71; 0.66]	Editor-level	
0.15 [0.01; 0.29]	0.17 [-0.16; 0.51]	0.19 [-0.40; 0.79]	Control

Table S8. Netsplit for the different peer-review interventions for peer-review duration (random effects model). Example: 12 studies compared standard peer-review process vs reviewer-level interventions and the estimated treatment effect (standardized mean difference [SMD]) was -0.1481.

Random effects model:

comparison	k	prop	nma	direct	indir.	Diff	z	p-value
Author-level:Control	2	1.00	0.1710	0.1710	.	.	.	.
Author-level:Editor-level	0	0	-0.0234	.	-0.0234	.	.	.
Author-level:Reviewer-level	0	0	0.0228	.	0.0228	.	.	.
Control:Editor-level	1	1.00	-0.1943	-0.1943	.	.	.	.
Control:Reviewer-level	12	1.00	-0.1481	-0.1481	.	.	.	.
Editor-level:Reviewer-level	0	0	0.0462	.	0.0462	.	.	.

Legend:
- comparison - Treatment comparison
- k - Number of studies providing direct evidence
- prop - Direct evidence proportion
- nma - Estimated treatment effect (SMD) in network meta-analysis
- direct - Estimated treatment effect (SMD) derived from direct evidence
- indir. - Estimated treatment effect (SMD) derived from indirect evidence
- Diff - Difference between direct and indirect treatment estimates
- z - z-value of test for disagreement (direct versus indirect)
- p-value - p-value of test for disagreement (direct versus indirect)

Table S9. Quantifying heterogeneity/inconsistency, tests of heterogeneity (within designs) and inconsistency (between designs), and design-specific decomposition of within-designs Q statistic (peer-review duration).

```
Q statistics to assess homogeneity / consistency

                     Q df  p-value
Total            60.62 12 < 0.0001
Within designs   60.62 12 < 0.0001
Between designs   0.00  0       --

Design-specific decomposition of within-designs Q statistic

                  Design     Q df  p-value
   Author-level:Control  4.46  1   0.0347
 Control:Reviewer-level 56.16 11 < 0.0001

Q statistic to assess consistency under the assumption of
a full design-by-treatment interaction random effects model

                     Q df p-value tau.within tau2.within
Between designs -0.00  0      --     0.2184      0.0477
```

Table S10. League table for the duration of the peer-review process (fixed effect model).

Reviewer-level			
0.01 [-0.13; 0.16]	Author-level		
-0.07 [-0.49; 0.36]	-0.08 [-0.52; 0.36]	Editor-level	
0.13 [0.07; 0.19]	0.12 [-0.02; 0.25]	0.19 [-0.22; 0.61]	Control

Table S11. Netsplit for the different peer-review interventions for peer-review duration (fixed effect model). Example: 12 studies compared standard peer-review process vs reviewer-level interventions and the estimated treatment effect (standardized mean difference [SMD]) was -0.1293.

Fixed effect model:

comparison	k	prop	nma	direct	indir.	Diff	z	p-value
Author-level:Control	2	1.00	0.1162	0.1162	.	.	.	.
Author-level:Editor-level	0	0	-0.0781	.	-0.0781	.	.	.
Author-level:Reviewer-level	0	0	-0.0131	.	-0.0131	.	.	.
Control:Editor-level	1	1.00	-0.1943	-0.1943	.	.	.	.
Control:Reviewer-level	12	1.00	-0.1293	-0.1293	.	.	.	.
Editor-level:Reviewer-level	0	0	0.0650	.	0.0650	.	.	.

Legend:
- comparison - Treatment comparison
- k - Number of studies providing direct evidence
- prop - Direct evidence proportion
- nma - Estimated treatment effect (SMD) in network meta-analysis
- direct - Estimated treatment effect (SMD) derived from direct evidence
- indir. - Estimated treatment effect (SMD) derived from indirect evidence
- Diff - Difference between direct and indirect treatment estimates
- z - z-value of test for disagreement (direct versus indirect)
- p-value - p-value of test for disagreement (direct versus indirect)

Table S12. League table for the duration of the peer-review process (reviewer-level interventions; random effects model).

Blinding			
-0.28 [-0.62; 0.06]	Training		
-0.44 [-0.74; -0.15]	-0.17 [-0.54; 0.20]	Other_intervention	
0.01 [-0.17; 0.19]	0.28 [0.00; 0.57]	0.45 [0.21; 0.68]	Control

Table S13. Netsplit for the different peer-review interventions for peer-review duration (reviewer-level interventions, random effects model). Example: 8 studies compared blinding vs standard peer-review process and the estimated treatment effect (standardized mean difference [SMD]) was 0.0069.

Random effects model:

comparison	k	prop	nma	direct	indir.	Diff	z	p-value
Blinding:Control	8	1.00	0.0069	0.0069	.	.	.	.
Blinding:Other_intervention	0	0	-0.4422	.	-0.4422	.	.	.
Blinding:Training	0	0	-0.2767	.	-0.2767	.	.	.
Control:Other_intervention	6	1.00	-0.4491	-0.4491	.	.	.	.
Control:Training	5	1.00	-0.2836	-0.2836	.	.	.	.
Other_intervention:Training	0	0	0.1654	.	0.1654	.	.	.

Legend:

comparison	Treatment comparison
k	Number of studies providing direct evidence
prop	Direct evidence proportion
nma	Estimated treatment effect (SMD) in network meta-analysis
direct	Estimated treatment effect (SMD) derived from direct evidence
indir.	Estimated treatment effect (SMD) derived from indirect evidence
Diff	Difference between direct and indirect treatment estimates
z	z-value of test for disagreement (direct versus indirect)
p-value	p-value of test for disagreement (direct versus indirect)

Table S14. Quantifying heterogeneity/inconsistency, tests of heterogeneity (within designs) and inconsistency (between designs), and design-specific decomposition of within-designs Q statistic (peer-review duration – reviewer-level interventions).

```
Q statistics to assess homogeneity / consistency

                     Q df  p-value
Total            67.72 16 < 0.0001
Within designs   67.72 16 < 0.0001
Between designs -0.00  0       --

Design-specific decomposition of within-designs Q statistic

                      Design     Q df  p-value
           Blinding:Control 14.23  7   0.0472
 Control:Other_intervention 51.20  5 < 0.0001
           Control:Training  2.29  4   0.6827

Q statistic to assess consistency under the assumption of
a full design-by-treatment interaction random effects model

                   Q df p-value tau.within tau2.within
Between designs 0.00  0      --     0.2338      0.0547
```

Figure S1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart of our analysis.

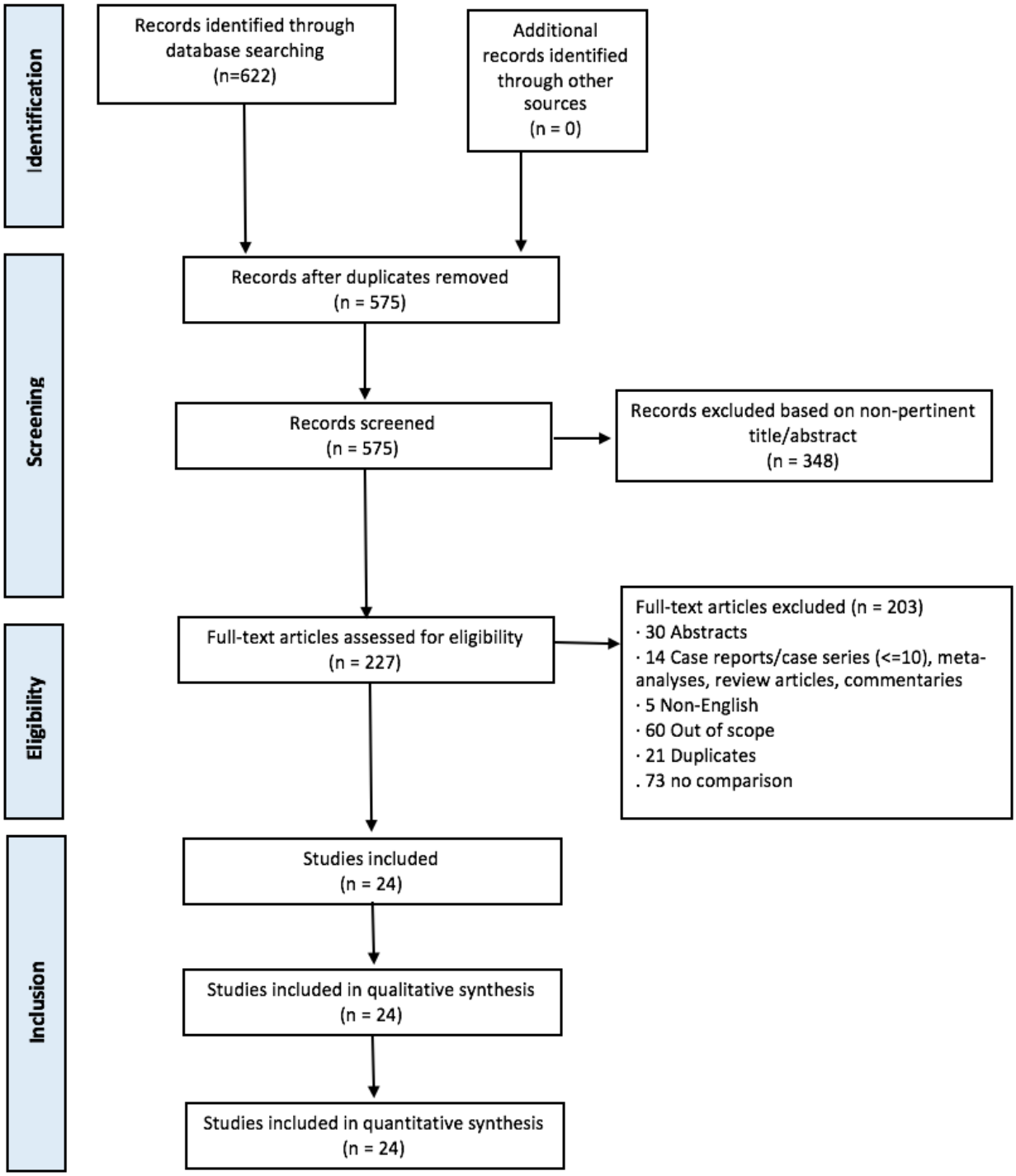


Figure S2. **Net graph for the main outcome of peer-review quality.**

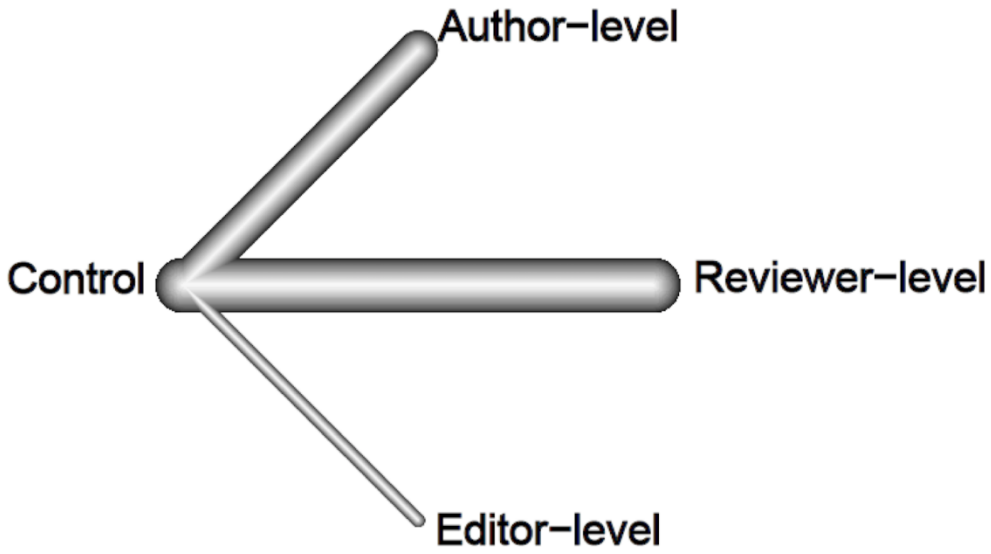


Figure S3. Funnel plot for the assessment of publication bias for the main outcome of peer-review quality.

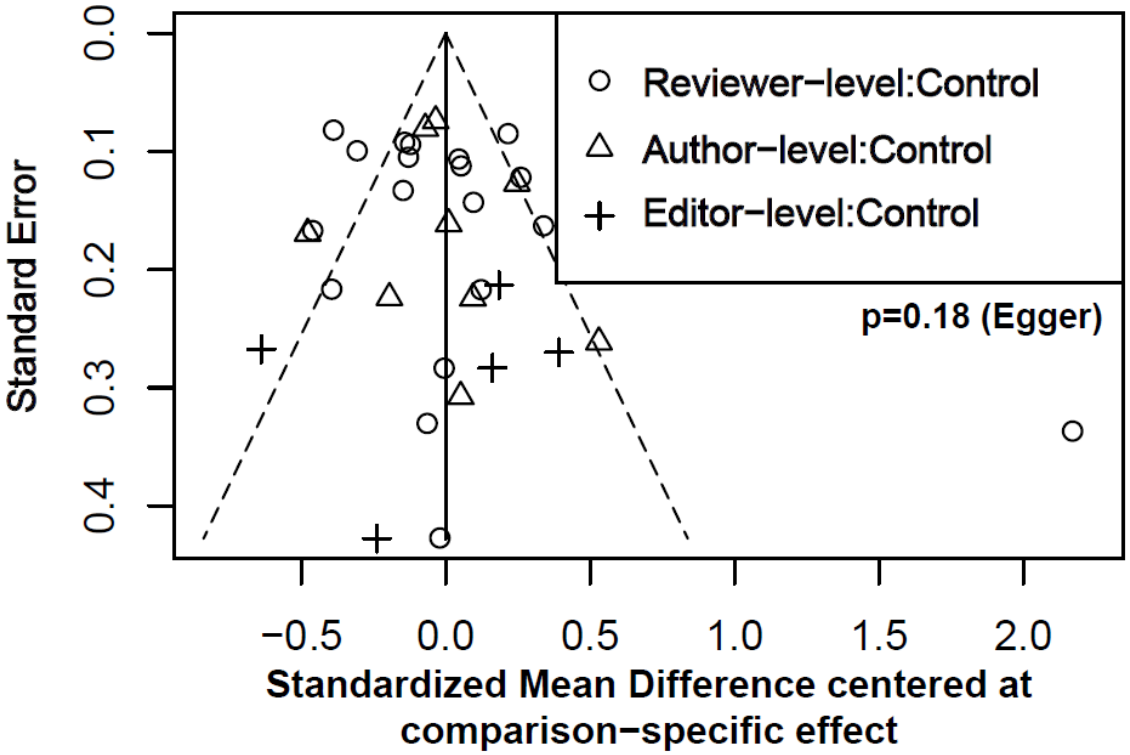


Figure S4. Leave-one-out analysis for standardized mean difference for the main outcome of peer-review quality (random effects model).

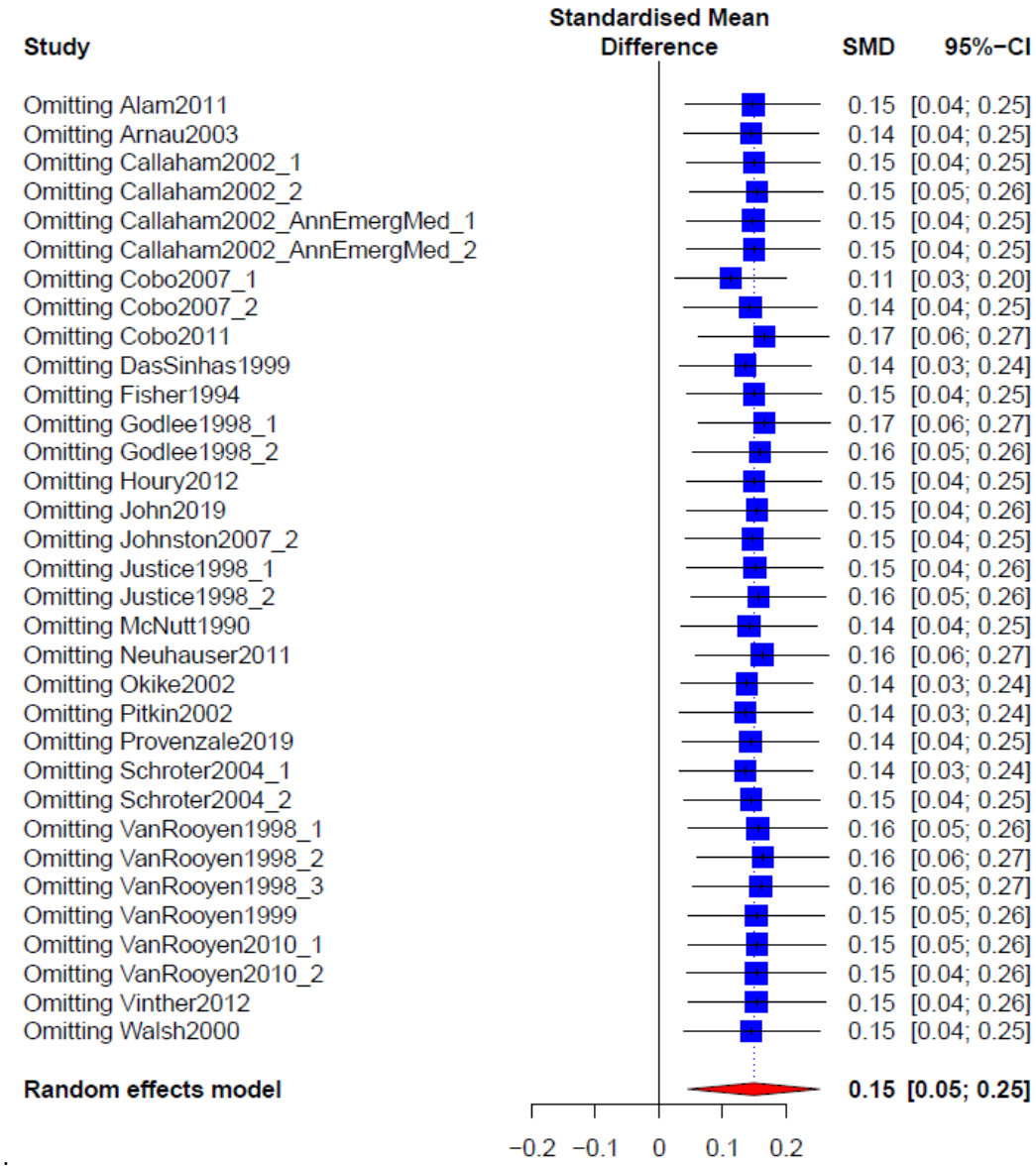
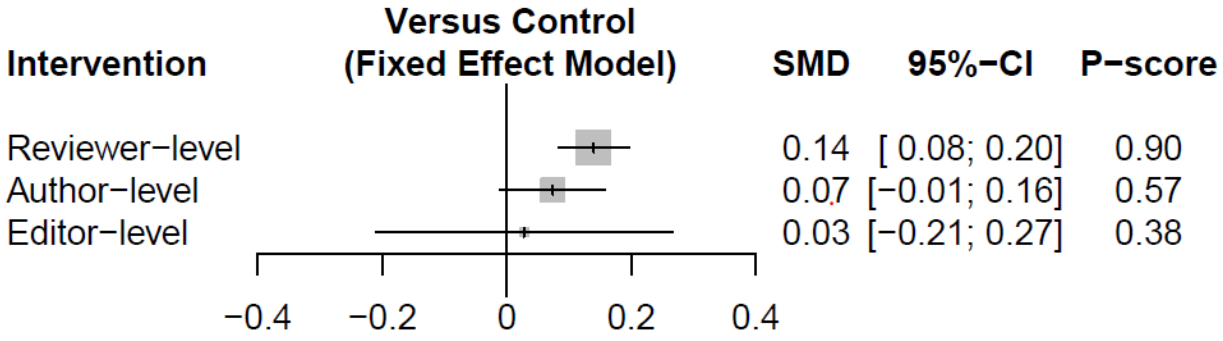


Figure S5. Network forest plot for quality of the peer-review process among the different interventions (fixed effect model).



Larger p-scores signify larger standardized mean difference (SMD) vs control and larger intervention effect on peer-review quality.

Figure S6. Leave-one-out analysis for standardized mean difference for the main outcome of peer-review quality (fixed effect model).

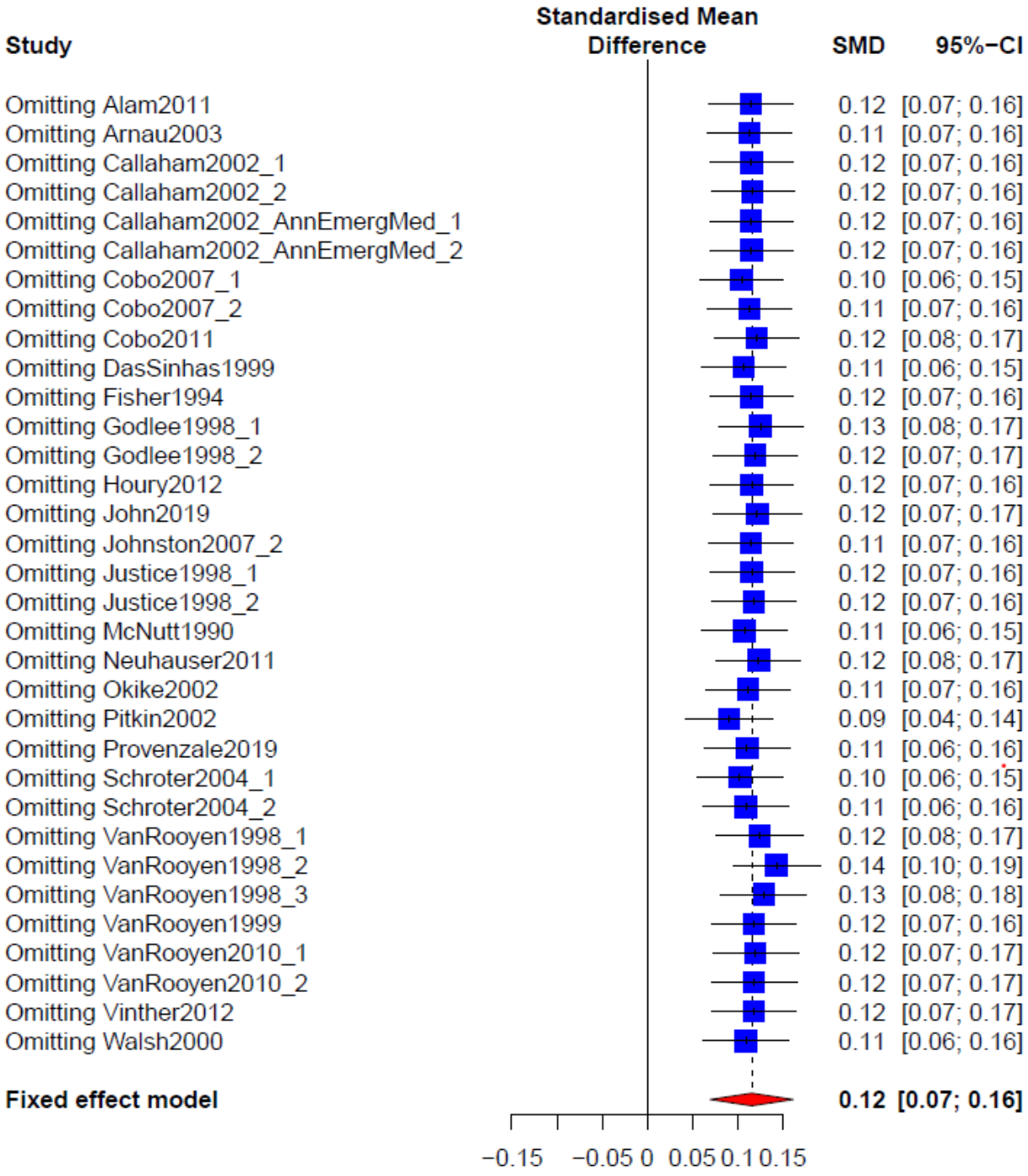


Figure S7. **Net graph for the duration of the peer-review process.**

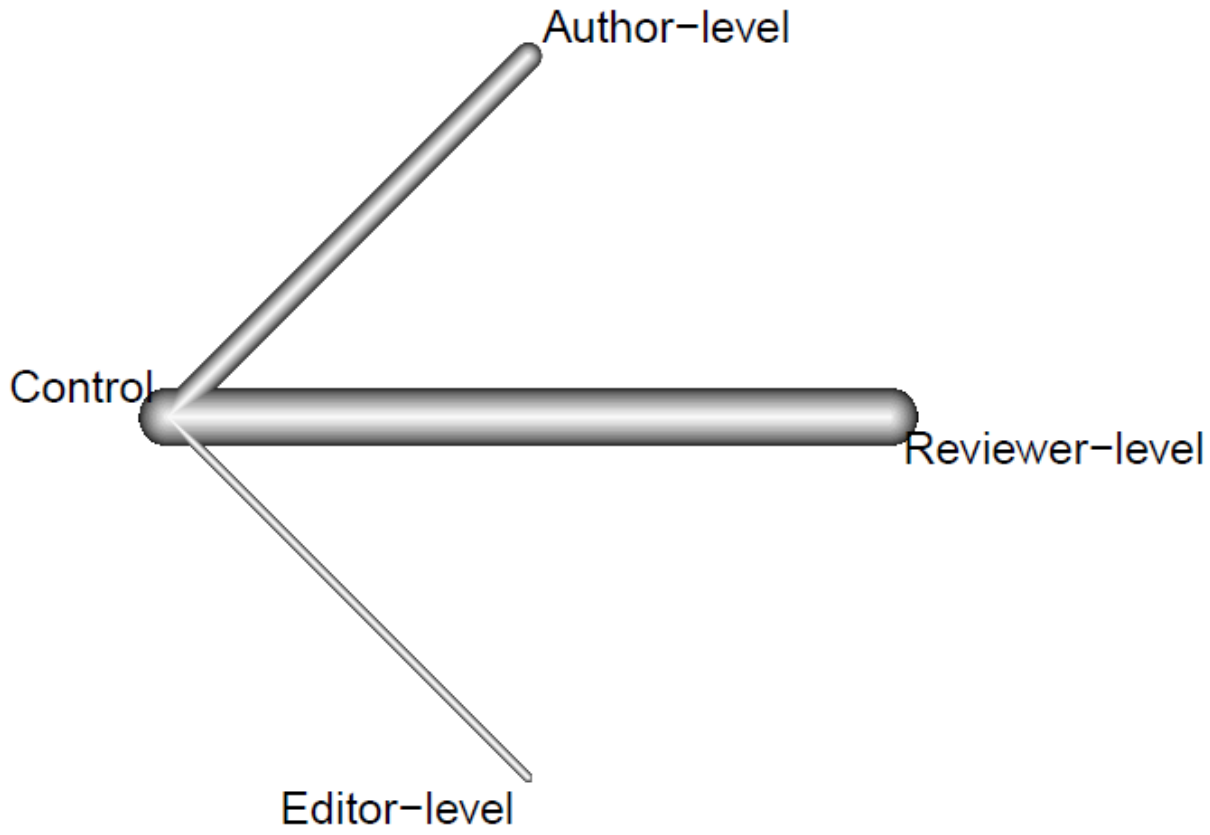


Figure S8. Funnel plot for the assessment of publication bias for duration of the peer-review process.

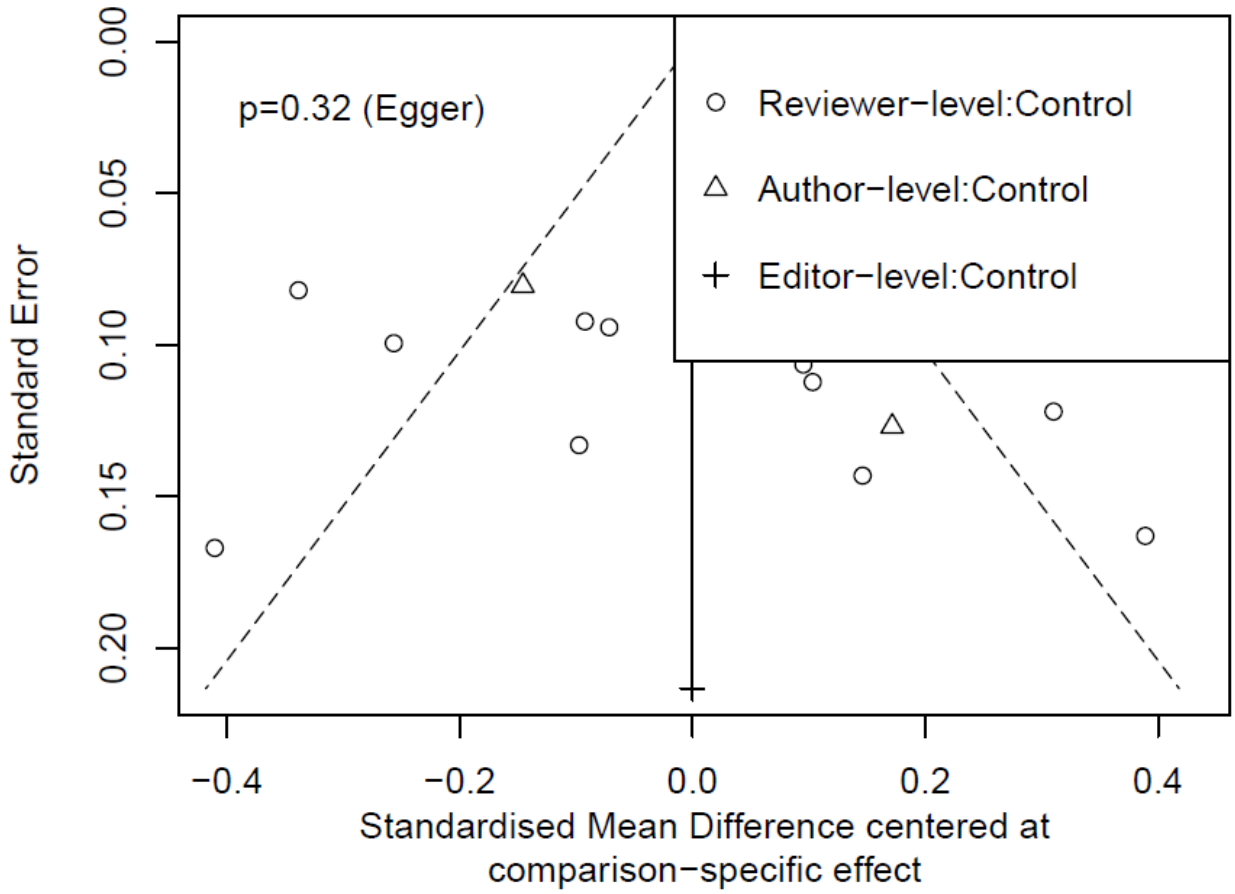


Figure S9. Leave-one-out analysis for standardized mean difference for peer-review duration (random effects model).

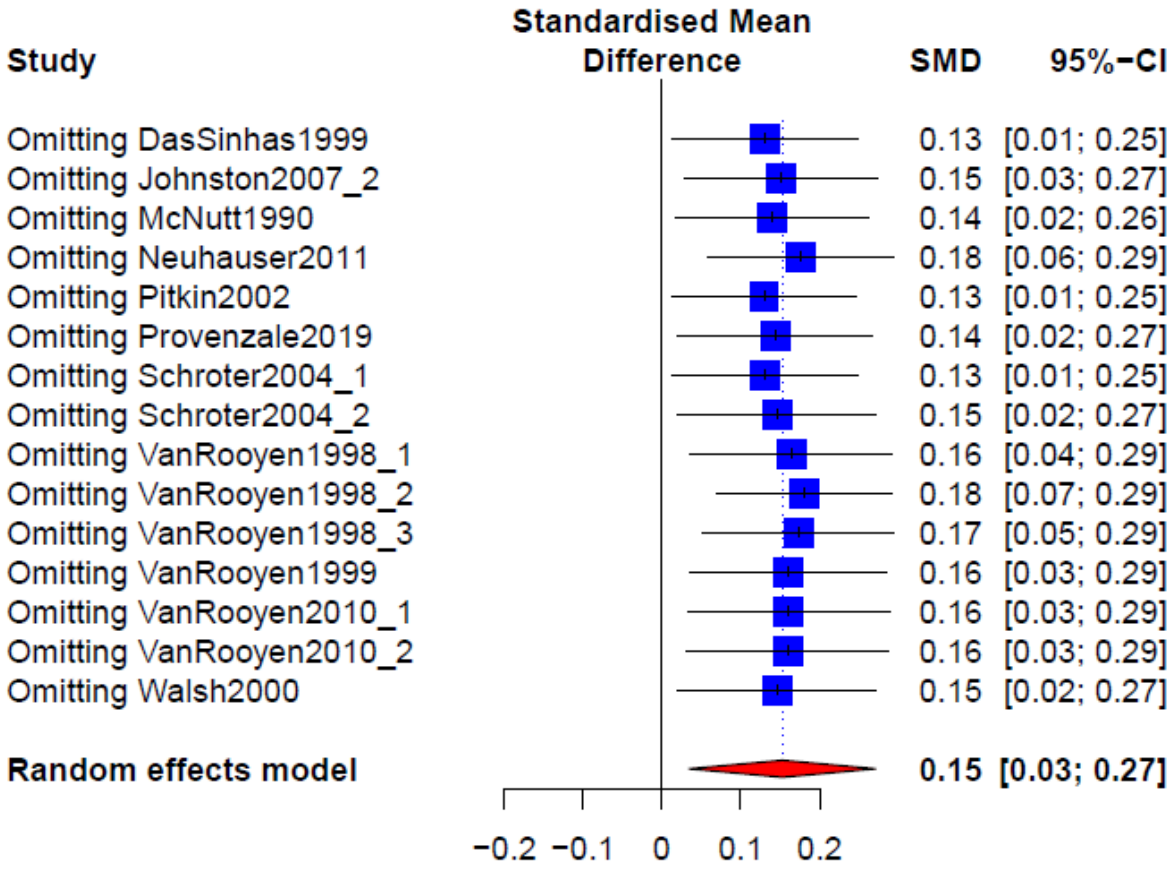
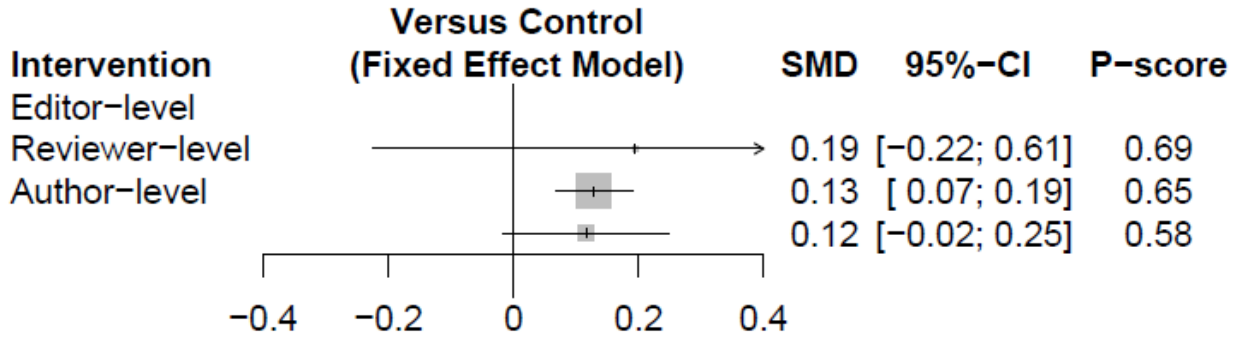


Figure S10. Network forest plot for peer-review duration among the different interventions (fixed effect model).



Larger p-scores signify larger standardized mean difference (SMD) vs control and larger intervention effect on peer-review quality.

Figure S11. Leave-one-out analysis for standardized mean difference for peer-review duration (fixed effect model).

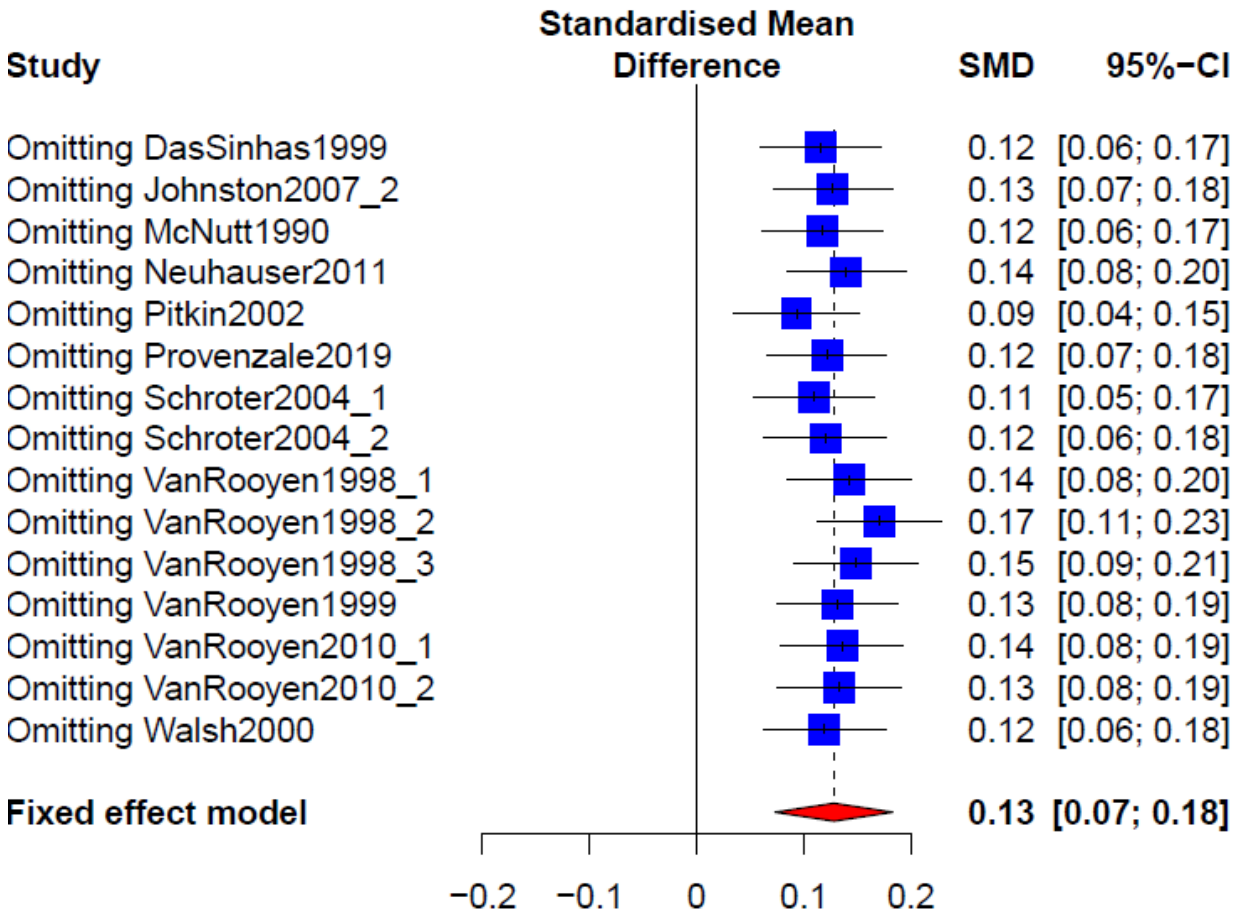
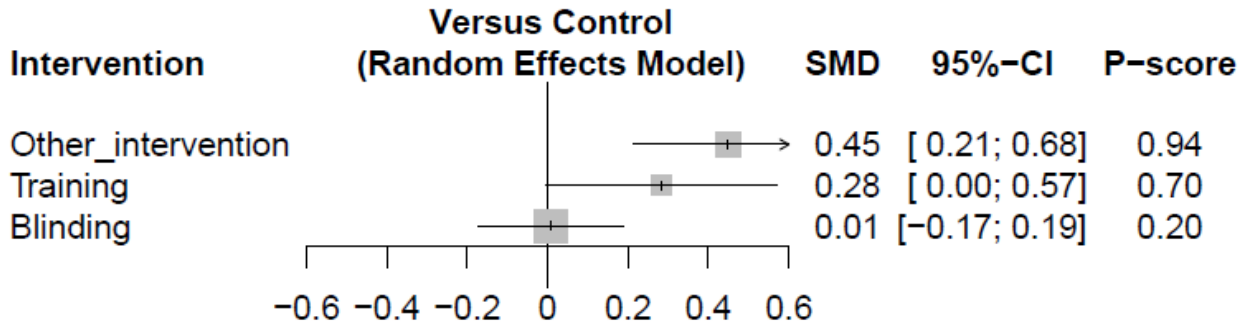


Figure S12. **Network forest plot for peer-review duration among the different reviewer-level interventions (random effects model).**



Larger p-scores signify larger standardized mean difference (SMD) vs control and larger intervention effect on peer-review duration.

Figure S13. **Net graph for the duration of the peer-review process (reviewer-level interventions).**

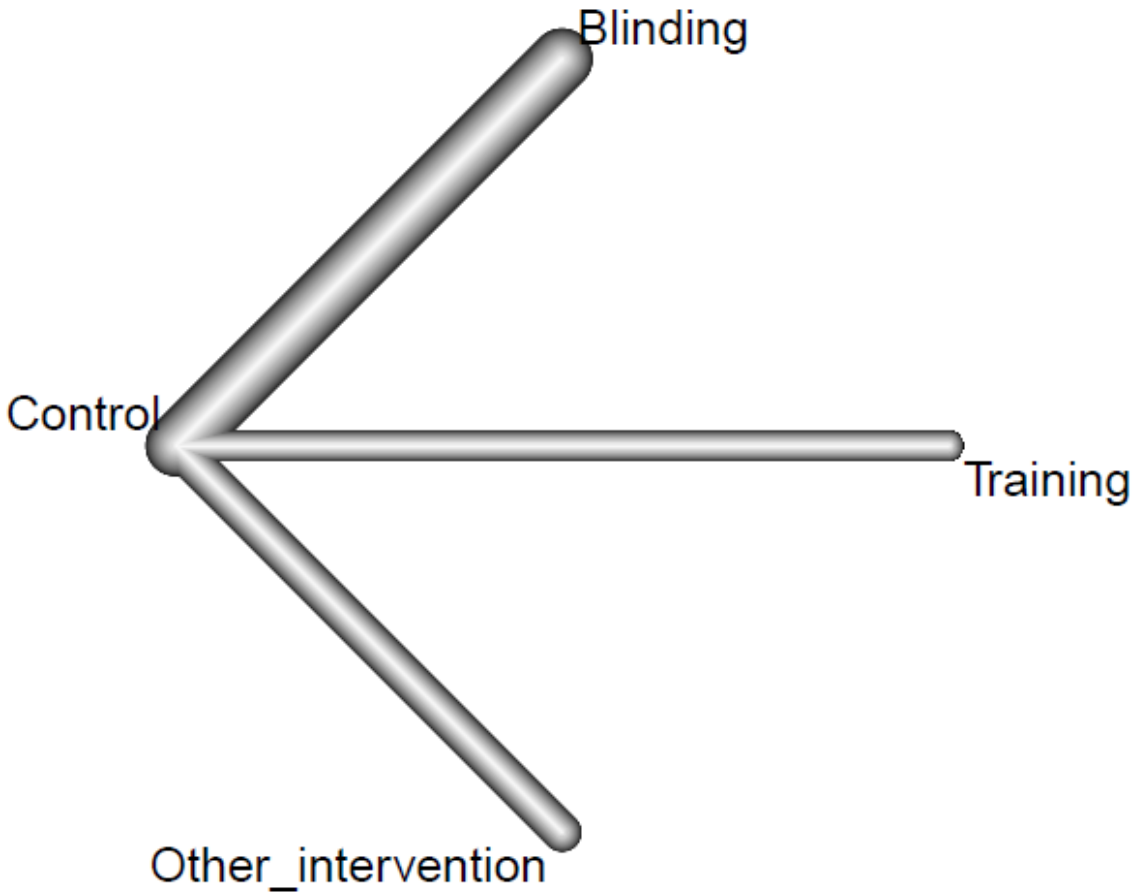


Figure S14. Funnel plot for the assessment of publication bias for duration of the peer-review process (reviewer-level interventions).

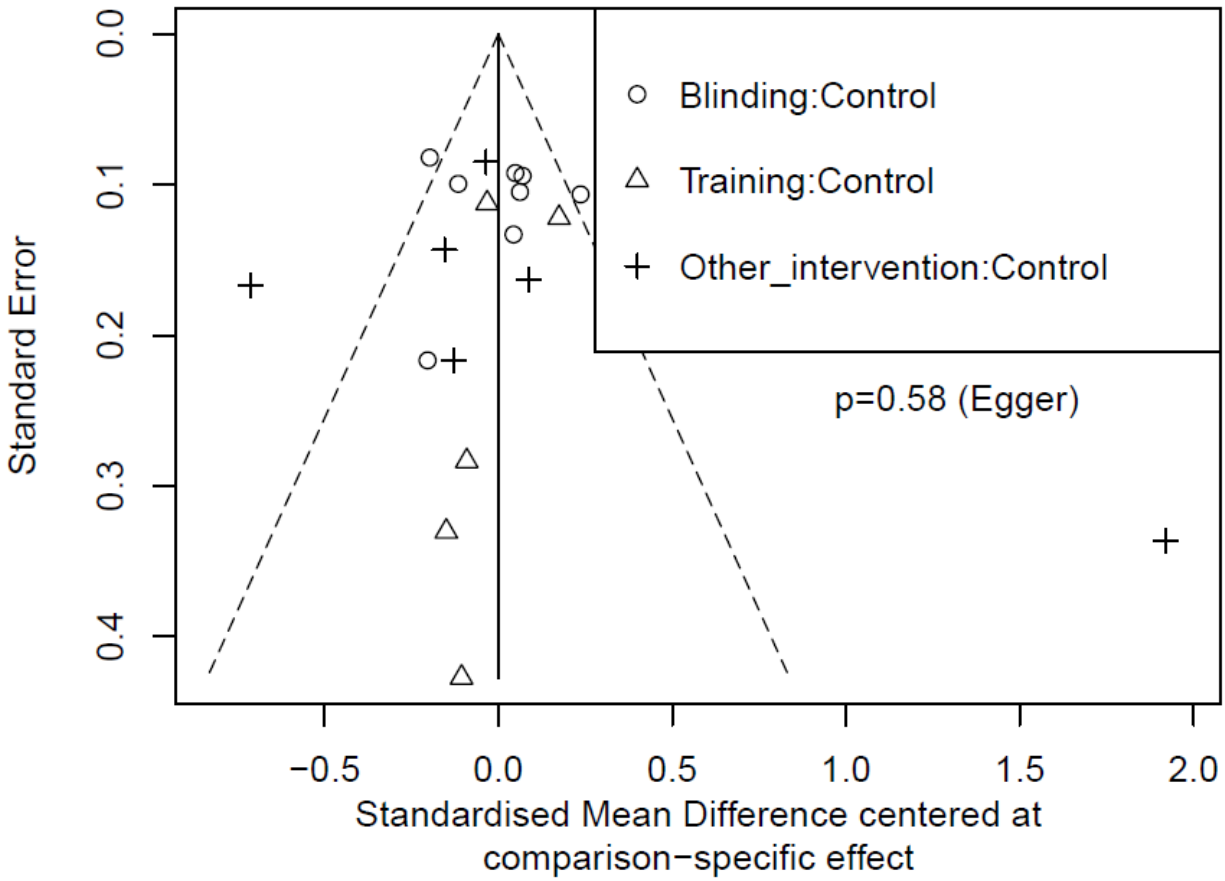


Figure S15. Leave-one-out analysis for standardized mean difference for peer-review duration (reviewer-level interventions; random effects model).

